***What explains the variability in conflict-related sexual violence across countries?***

## I. Introduction:

During the Rwandan Genocide, an estimated 100,000 to 250,000 women were raped within just three months (UN). Similarly, during the war in Yugoslavia, more than 20,000 women were subjected to sexual violence (Amnesty International). Conflict-related sexual violence (CRSV) is one of the most harrowing and pervasive aspects of modern warfare, inflicting unimaginable suffering and leaving long-lasting scars. Survivors often grapple with severe consequences, including HIV/AIDS, profound psychological trauma, social stigma, and economic marginalization. Beyond the individual, these acts disrupt families, destabilize communities, and hinder post-conflict recovery, perpetuating cycles of poverty, violence, and social fragmentation(Amnesty International). CRSV is not just a consequence of war but a weapon that amplifies its devastation.

The conceptual definition of CRSV, is defined by the United Nations Secretary General Special Representation on Conflict Related Sexual Violence is : " rape, sexual slavery, forced prostitution, forced pregnancy, forced abortion, enforced sterilization, forced marriage, and any other form of sexual violence of comparable gravity perpetrated against women, men, girls or boys that is directly or indirectly linked to a conflict" (UNSGSR). These crimes are frequently used as tools of war to achieve strategic objectives: displacing populations, terrorizing communities, asserting control, or committing genocide by erasing future generations of an opposing group. Although it is widely recognized that CRSV occurs in most armed conflicts, its prevalence and severity vary greatly. A critical and unanswered question remains: Why do some conflicts experience extremely high levels of CRSV while others report far fewer or even no cases?

For example, widespread sexual violence characterized conflicts such as the genocide against the Tutsi in Rwanda, the Yugoslav Wars, the First and Second Congo Wars, and the Tigray War in Ethiopia (Cohen). These atrocities are starkly contrasted with conflicts like the El Salvador Civil War or those involving the Liberation Tigers of Tamil Eelam (LTTE) in Sri Lanka, where sexual violence was far less prevalent

(Cohen et al.). This variation raises pressing questions about the underlying drivers and conditions that influence the occurrence of CRSV. Are these differences rooted in cultural factors, the structure of armed groups, leadership strategies, or something else entirely?

Understanding this variation is vital. Identifying the factors that make certain conflicts more prone to CRSV can help policymakers and international organizations design targeted interventions to prevent its occurrence. Insights from this research could guide the United Nations and peacekeeping missions in deploying specialized task forces to protect vulnerable populations, especially women and children, in high-risk zones. Furthermore, by addressing the root causes of CRSV, global responses can shift from reactionary measures to proactive strategies, ultimately reducing the devastating human toll of these crimes.

Thus, my research seeks to address the following question: **What explains the variability in conflict-related sexual violence across conflicts?**

## II. Literature review:

The study of sexual violence during conflict reveals significant variability in its prevalence and underlying causes. Elisabeth Jean Wood, in *Variation in Sexual Violence during War*, examines why sexual violence varies greatly across conflicts and among armed groups. She attributes this variation to strategic decisions by leadership, norms within combat units, small group dynamics, and the effectiveness of military discipline. Wood posits that armed groups with robust command structures and norms condemning sexual violence are less likely to perpetrate it, while poorly disciplined groups or those whose leadership condones sexual violence are more prone to such actions. Her qualitative analysis, grounded in case studies from contexts such as Nazi Germany, Colombia, Bosnia-Herzegovina, and Rwanda, underscores the importance of norms, discipline, and the presence of female combatants in influencing these outcomes. For instance, insurgent groups that rely on civilian support often maintain strict discipline, reducing the likelihood of sexual violence. However, while Wood highlights key factors

that mitigate or exacerbate conflict-related sexual violence (CRSV), her reliance on qualitative data and a limited sample size raises questions about the generalizability of her findings. There is a notable absence of large-scale quantitative analysis, leaving unanswered questions about the relative weight of these factors in explaining CRSV. This gap underscores the need for broader datasets and mixed-methods approaches to validate her conclusions on a global scale.

Christopher K. Butler and Jessica L. Jones take a quantitative approach in their study, *Sexual Violence by Government Security Forces: Are Levels of Sexual Violence in Peacetime Predictive of Those in Civil Conflict?*. They challenge the assumption that conflict inherently increases sexual violence, arguing instead that pre-conflict levels of sexual violence are better predictors of its prevalence during conflict. Using data from 170 countries between 1999 and 2011, their study evaluates the relationship between peacetime and conflict-period sexual violence committed by government security forces. Their analysis employs a five-point scale measuring sexual violence severity and uses statistical methods such as ordered probit regressions and cross-tabulations. The findings indicate that countries with high levels of sexual violence in the years leading up to conflict are likely to sustain these levels during conflict, suggesting that societal norms and pre-existing patterns of sexual violence play a significant role. While this research provides valuable insights, its narrow focus on state actors excludes other key perpetrators, such as rebel groups and militias, which limits the applicability of its findings to a broader understanding of CRSV. Moreover, the study does not account for how societal beliefs and attitudes toward sexual violence might shape its occurrence during conflict, raising questions about whether state actors' behavior adequately represents overall trends in CRSV. Addressing these gaps requires a more comprehensive dataset that includes diverse perpetrators and a consideration of societal norms.

Dara Kay Cohen's research, *Explaining Rape during Civil War: Cross-National Evidence (1980–2009)*, investigates the variation in sexual violence committed by armed actors during civil wars, emphasizing the role of recruitment methods. Cohen introduces the combatant socialization hypothesis, arguing that groups employing forced recruitment methods, such as abduction or pressganging, are more likely to

commit rape, particularly gang rape, as a means of fostering unit cohesion. Her statistical analysis, based on an original dataset spanning civil wars from 1980 to 2009, supports this hypothesis. Specifically, her findings reveal that abduction is associated with insurgent-perpetrated rape, while pressganging predicts state-perpetrated rape. By contrast, weaker forms of forced recruitment, such as conscription, do not significantly correlate with wartime rape, highlighting the unique role of violent recruitment methods in driving such acts. Cohen also examines other hypotheses, including those related to opportunism, ethnic hatred, and gender inequality, but finds mixed results. For instance, while state collapse correlates with insurgent-perpetrated rape, neither ethnic hatred nor gender inequality significantly predicts wartime rape. These findings challenge prevailing assumptions about the centrality of gender inequality in explaining sexual violence during war, suggesting that its influence may be more relevant to conflict onset than to variations in wartime sexual violence. This insight points to the need for alternative measures of gender inequality, rather than just fertility rates, such as for example pre-conflict sexual violence rates, to better capture societal norms around consent and sexual violence.

Together, these studies contribute critical perspectives to the literature on CRSV, highlighting both its complex determinants and the limitations of existing research. Wood's qualitative case studies provide nuanced insights into the role of norms and leadership but lack the statistical rigor to generalize findings across conflicts. Butler quantitative approach establishes a foundational relationship between pre-conflict and conflict-period sexual violence but is constrained by its focus on state actors. Cohen's work bridges these gaps by introducing a novel hypothesis about recruitment mechanisms, though her findings reveal the need for further exploration of societal norms and their influence on CRSV. Collectively, these gaps suggest a pressing need for interdisciplinary research that combines qualitative and quantitative methods while expanding the scope of actors and contexts analyzed. Which has led my investigation to discover how peacetime sexual violence in countries predict conflict related sexual Violence:

## III. Data and Methods:

The hypothesis of this study is that countries with a high pre-conflict prevalence of sexual violence prior are more likely to experience high levels of conflict-related sexual violence (CRSV). This suggests that sexual violence during conflict is more likely to be committed by individuals from countries where sexual violence was already prevalent and probably normalized. That the extent of violations women experience before war (such as sexual violence) are driven by the same patriarchy and misogynistic thinkings, that drive the sexual abuse of women during conflict. This claim is grounded in the theory of the "continuum of violence", which was developed by Liz Kelly (Kelly, 1988) Dr. Kelly suggests a "theoretical change to the understanding of violence agaisnt women as episodic and deviant incidents (defined in law as crimes) of extreme cruelty and harm, to recognizing that it is normative and functional: an everyday context for the lives and experiences of women and girls all over the world" (Kelly, 1988). Applying the theory of the continuum of change to CRSV, would suggests that wartime sexual violence is not an isolated phenomenon, but rather exists as part of a continuum of gender-based violations against women that begin in peacetime and extend into wartime (Card 1996, Enloe 2000). Other feminist scholars such as Card and Enloe have expanded on this theoretical Framework, by arguing that wartime sexual violence is not only closely tied to pre-conflict sexual violence but also the political, social, and economic status of women in the prewar period, factors that legtimise and allow for sexual violence against women(Card 1996, Enloe 2000). This theory of the continuum of violence gives us the rationale to hypothesize that there is a positive relationship between countries pre-conflict sexual violence prevalence and their conflict related sexual Violence. If a country exhibits high levels of sexual violence in peacetime, it is more likely that these patterns of violence will continue and escalate during periods of conflict. Similarly, countries with low levels of sexual violence prior to conflict would be expected to experience lower levels of CRSV. Investigating this relationship and finding a relationship could potentially answer our research questions of what explains the variability in CRSV accross different conflicts.

To test this hypothesis, we used a specific Variable for the dependent variable (DV) and a specific independent variable (IV):

**III. ii Dependent Variable:**

The dependent variable in this study is the level of conflict related sexual Violence. Measured on an ordinal level of measurement of 0-4, the dependent variable is measured by a hierarchical categorisation of sexual violence committed by government security forces. This measure is adapted from Butler, Gluch, and Mitchell (2007) and the coding of the code on a five-point scale ranging from 0 to 4 rely on information obtained from the United States Department of State Human Rights Country Reports, (Butler, C. K., & Jones, J. L., 2016).The coding rubric from Butler et.all is presented in Table1.

Table 1: Coding Rubric for Sexual Violence Committed by Security Forces

| Level | Description |
|---|---|
| 0 | No mention of 'rape,' 'sexual assault,' or 'sexual abuse' by security forces. |
| 1 | Isolated or alleged reports of 'rape,' 'sexual assault,' or 'sexual abuse' by security forces. |
| 2 | 'Reports' or indications that SV occurred 'sometimes,' 'continued to be a problem,' or 'remained an issue.' |
| 3 | 'Numerous reports' of SV described as 'routine,' 'common,' 'widespread,' or 'systematic,' including its use as a tool of torture to extract information, intimidate, or punish targeted groups (e.g., detainees, opposition members, or ethnic minorities). |
| 4 | SV was used as 'a tool of war' or 'a systematic weapon of war.' |

According to Butler and et.all The term "security forces" includes militaries, government-supported militias, police, and prison guards, but excludes warlords, pro-government militias, and other non-state actors (Butler, C. K., & Jones, J. L, 2016). Importantly, this measure focuses on Sexual Violence committed against civilians, rather than combatants.

The unit of analysis for the dependent variable is the country-year, and the data cover 170 countries across a thirteen-year period (1999–2011), resulting in 2,207 observations. For countries experiencing multiple

years of conflict, the analysis selected the highest recorded CRSV score during the conflict period. This approach aligns with the hypothesis claims, that high pre-conflict prevalence of sexual violence predicts higher levels of conflict-related sexual violence, making it logical to use the peak intensity of CRSV during the conflict. Hence the operational definition of our dependent variable of (CRSV) is the (highest) level of conflict related sexual violence committed by government forces agaisnt civilians during conflict.

**III. ii. Independent Variable.**

The independent variable is the pre-conflict prevalence of Sexual Violence. It is measured on an interval level of measurement; and is measures the the Age-Standardized All Risk Factors Summary Exposure Value (SEV) for sexual violence, and it's values are log transformed. (called sev_sv in the dataset). This variable is derived from the Global Burden of Disease (GBD) dataset, and in simpler terms: The "Age-Standardized" part means that the data is adjusted to account for different age groups, so the value is a representation of the different age groups in the country's population. The "All Risk Factors" refers to the fact that the value takes into account different causes or conditions that contribute to sexual violence, and the "Summary Exposure Value" is a combined score that reflects the total exposure/prevalence to sexual violence in a given population (GBD). While SEV is commonly applied to assess exposure to diseases, given that SEV reflects the proportion of a population exposed to a particular risk factor, it can be interpreted as an indicator of prevalence for that exposure. Therefore, in the context where SEV is used to measure exposure to sexual violence, it can be considered a measure of its prevalence within the population. It is also worth noting that the values obtained from the GBD dataset are log-transformed, values on the scale of e. While this transformation is beneficial for reducing skewness and stabilizing variance, it complicates the interpretability of the values. However, the transformed values fall within a

range of 0.004906453 and 1.226277434. An interval where smaller values representing lower exposure to Sexual Violence and higher values represent high exposure to sexual violence.

Though the GBD offers the prevalence of sexual violence for each country for every year from 1980-2017. To make the variable pre conflict prevalence of sexual violence, the paper first had to identify the first year of conflict and later on the year before conflict. In order to do so, the GBD dataset was combined with the dataset from Butler et all (used for the dependent Variable) to determine the first year of conflict for each country. One year was subtracted from the first year of conflict to establish the pre-conflict year (in the dataset known as Year 0). After we located the log-transformed prevalence of conflict sexual violence for Year0 and called it the measured it the Pre-Conflict Prevalence of Sexual Violence" variable, for simplicity reasons. Like the dependent variable, the unit of analysis for the independent variable is country-year.

After combining the datasets, we obtained data for 130 countries. A few countries were excluded due to the inability to locate their pre-conflict years in the Butler dataset. For each country, the dataset includes the country name (Country)[1], the first year of conflict (first_year_of_conflict), the pre-conflict year (Year0), the level of CRSV (statedeptsvcode), and the pre-conflict prevalence of sexual violence (sev_sv). In total, the dataset contains 230 entries, as it includes data for both the male and female populations across all 130 countries.

## IV. Analysis and Results:

### IV. i Bivariate analysis between variables:

*Hypothesis:* Countries with high pre-confict prevalence of sexual violence will have high levels of CRSV.

*Null Hypothesis*: There is no relationship between Pre-conflict prevalence of Sexual Violence and the level CRSV.

[1] The terms in parentheses indicate how the variables are defined in the dataset.

To test our hypothesis, this paper will conduct an a bivariate Anova test to test the relationship between a country's  pre-conflict prevalence of sexual violence and the contry's level of conflict related sexual violence. An anova test was chosen as the type of statistical test to use as ANOVA compares one interval variable (preconflict prevalence of Sexaul violence) with an ordinal variable with more than 2 categories (level of CRSV), which is suitable in our case as our dependent variable is an ordinal scale from 0-4.  The Hypothesis test summaries are presented in Table2 below presents the mean values of across different CRSV levels

*Table2: Anova Table of the average Pre-conflict Sexual violence prevalence for each level of  CRSV.*

| **Level of CRSV** | **Average sev_sv** |
|---|---|
| 0 | 0.3703839 |
| 1 | 0.4180314 |
| 2 | 0.3057429 |
| 3 | 0.3137802 |
| 4 | 0.3592940 |

*Table 2: Summary statistics of the results of the Anova test, where level of CRSV represents the ordinal levels of CRSV (from 0-4) and Average sev_sv represent the anova average values of the pre-conflict prevalence of sexual violence.*

| Variable | Eta Squared | Partial Eta Squared |
|---|---|---|
| sev_sv | 0.007094883 | 0.007094883 |

Table 3: Eta Squared Results

When comparing the means, there is no clear, consistent trend between CRSV levels and the mean pre-conflict prevalence of sexual violence. Specifically, there is a sharp increase from score 0 (mean = 0.37) to score 1 (mean = 0.41), which suggests a notable difference between countries with no reported CRSV (score 0) and countries with isolated cases or reports of CRSV (score 1). However, this increase is not consistent across all levels. After score 1, the means begin to decrease. The mean for score 2 is 0.305, which is lower than score 1, and this contradicts our hypothesis that higher CRSV would correlate with higher levels of pre-conflict sexual violence.

After level 2, the mean increases slightly at level 3 (0.31), and further increases to 0.36 at level 4. These increases after level 2 seem to support our hypothesis that higher CRSV levels are associated with higher pre-conflict sexual violence. However, the decrease from score 1 to score 2 challenges the hypothesis of a consistent positive relationship between levels of CRSV and pre-conflict sexual violence. Therefore, there is no clear linear relationship between CRSV levels and the pre-conflict prevalence of sexual violence.

The eta-squared value of 0.007094883 indicates a very small effect size, suggesting a weak relationship between the two variables. Eta-squared values close to 0.01 are typically considered small, reinforcing that the impact of pre-conflict sexual violence on CRSV is minimal. Moreover, the p-value of 0.315 is much higher than the 0.05 significance threshold, which means the ANOVA test fails to reject the null hypothesis. This implies there is no statistically significant relationship between the two variables.

However, the sharp difference in mean sev_sv values between levels 1 and 2 suggests that there is a substantial distinction between countries with little to no sexual violence (score 1) and those with some sexual violence (score 2). This discrepancy might indicate an opportunity to categorize CRSV scores into two groups: low CRSV (scores 0 and 1) and high CRSV (scores 2, 3, and 4). This categorization could lead to a more statistically significant analysis, potentially supporting our hypothesis. By transforming the dependent variable into a nominal scale (low and high CRSV), we could run a logistic regression analysis, which may provide a clearer understanding of the relationship between the variables.

*Logistic regression:*

Y0: Low  CRSV (Combination of 0&1 levels of CRSV)
Y1: High CRSV (Combination of 2&3&4 levels of CRSV)

The paper will conduct a logistic regression analysis to examine the relationship between the independent variable, pre-conflict sexual violence (denoted as *sev_sv* in our dataset), and the dependent variable, levels of conflict-related sexual violence (CRSV), represented by *statedeptsv_code* in our dataset.

| Variable | Estimate | Std. Error | z value | p-values |
|---|---|---|---|---|
| (Intercept) | 2.6289 | 0.5086 | 5.1689 | $2.36 \times 10^{-7}$ |
| sev_sv | -0.6657 | 1.1187 | -0.5950 | 0.5518 |

Table 2: Logistical Regression Results:Coefficient Estimates, Standard Errors, z Values, and p-Values

To interpret the relationship provided by the logistical regression, examining the coefficient estibate of preconfict prevalence of sexual violence (sev_sv) is of interest to our research. The coefficient for sev_sv is -0.6657. This coefficient suggests a negative relationship between pre-conflict sexual violence (sev_sv) and the likelihood of experiencing high CRSV. In other words, for every additional unit of sev_sv, the odds of moving from low CRSV (Y0) to high CRSV (Y1) are roughly approximately 51.4% (calculated as e^(-0.6657) = 0.514), In other words, the odds of moving from a "low CRSV" category (Y0) to a "high CRSV" category (Y1) decrease by approximately 48% (1 - 0.514 = 0.486) for every additional unit of pre-conflict prevalence of sexual violence. This suggests a negative relationship between the pre-conflict sexual violence prevalence and the occurrence of CRSV, implying that higher pre-conflict sexual violence is associated with a reduced likelihood of high CRSV. Which goes against our original hypothesis that higher pre-conflict sexual violence would lead to a higher levels of CRSV. Additionally, the relationship seems to be weak as it has a low McFadden's $R^2$, (which is a common measure of fit in logistic regression) of 0.0041. This value[2] indicates a very weak relationship between the IV and DV. And also it

[2] Appendix Code 11: result from Bivariate Analysis for logistic regression

has high p-value of 0.5518, which means this statistical test fails to reject the null hypothesis that there is no relationship between countries pre-connflict sexual violence and levels of CRSV.

***IV.ii Multivariate Analysis:***

The GBD dataset provides separate preconflict prevalence of sexual violence values for both male and female populations in each country, offering insights into pre-conflict sexual violence prevalence for each gender. This gender-specific data raises the question: *What happens when we control for sex?* Since males are generally known to experience lower levels of CRSV, the male sev_sv might be disrupting the linear relationship between the overall pre-conflict sexual violence and levels CRSV within countries. By controlling for sex, we may better understand the relationship between pre-conflict sexual violence and CRSV, potentially lending support to our hypothesis. To do so, we ran a logistic regression of high levels of CRSV (Y = 1) and low levels of CRSV (Y = 0) versus sex (Male = 1, Female = 0) versus pre-conflict prevalence of Sexual Violence (sev_sv).

| Variable | Estimate | Std. Error | z value | p-values |
|---|---|---|---|---|
| (Intercept) | 3.0624 | 0.9640 | 3.1768 | 0.0015 |
| sev_sv | -1.2497 | 1.5353 | -0.8140 | 0.4156 |
| sex_labelMale | -0.4618 | 0.8548 | -0.5403 | 0.5890 |

Table 3: Regression Results 2: Estimates, Standard Errors, z Values, and p-Values

This logistic regression model tests the relationship between pre-conflict prevalence of sexual violence (sev_sv) and the likelihood of transitioning from low to high levels of CRSV (as indicated by Y = 0 for low CRSV and Y = 1 for high CRSV) while controlling for gender. The odds ratio for preconflict prevalence of sexual violence is estimated to be e^(-1.2497) ≈ 0.287. This suggests that for every additional unit of sev_sv, the odds of experiencing high CRSV (vs. low CRSV) decrease by approximately 71.3%. This result indicates a negative relationship between pre-conflict sexual violence and CRSV after controlling for gender. Which once again contradicts our hypothesis. However when we conduct the McFadden's Pseudo R^2 which specifically measures the strength of the relationship between

variables. It indicates that its R^2 value is 0.0077, which value[3] is extremely small, which suggests that though there seems to be a negative relationship  between preconflict prevalence and conflict related sexual violence when controlling for sex, the relationship is extremely week

Furthermore, we observe a negative relationship between being male and the level of CRSV when controlling for pre-conflict prevalence of sexual violence. The coefficient for sex_labelMale is negative (-0.4618), indicating that, after controlling for preconflict prevalence of sexual violence, males tend to experience lower levels of CRSV compared to females. This aligns with expectations, as it is expected that women are more likely to experience sexual violence during conflict.

However, when revisiting our hypothesis, we originally expected a positive relationship between pre-conflict sexual violence and CRSV, even after controlling for sex. Instead, the regression results suggest an even stronger negative relationship between these variables, even after the control variable.

When examining statistical significance, It is also important to note that the p-values for both coefficients, sev_sv ($p = 0.4156$) and sex_labelMale ($p = 0.5890$), are greater than the standard significance threshold of 0.05. This indicates that the relationship between pre-conflict prevalence of sexual violence and CRSV is not statistically significant in this model. In other words, we fail to reject the null hypothesis, which suggests that there is no significant relationship between these variables when controlling for gender. Thus, while the model shows a negative relationship between sev_sv and CRSV and between sex and CRSV, the lack of statistical significance means we cannot confidently claim a meaningful association between pre-conflict sexual violence and CRSV in this dataset, even after considering sex as a factor.

## V. Discussions and Conclusions

Our bivariate and multivariate analyses both challenge the original hypothesis, which suggested that the pre-conflict prevalence of sexual violence is positively correlated with high levels of conflict-related

[3] Appendix code 12: Multivariate analysis

sexual violence (CRSV) as both analyses failed to reject the null hypothesis. However, the high p-values observed may be attributed to the small sample size of 130 countries. If we had more data points, we might have observed statistical significance, which could have supported the relationships suggested by the bivariate and multivariate analyses.

The first bivariate analysis run by an anova test suggested that there is no clear linear relationship between pre-conflict sexual violence and CRSV. However, it did reveal some interesting patterns. It showed that there was an increase in CRSV levels from most jumps, except from level 1 to level 2. This discrepancy could suggest that different hypotheses may be needed to explain the differences between differing levels, and that there are distinctions among the levels themselves. This insight adds to existing literature, which has primarily focused on an analysis of change in CRSV in a straightforward progression at an ordinary level of measurement, (Butler in particular from levels 0 to 4, and Cohen from Level 0-3), rather than examining the differences, patterns, and relationships between the different jumps of levels . These interlevel differences could potentially explain the intricate variability in CRSV among different conflicts, and what the intralevel similarities among those countries' conflicts are. However, as this relationship was statistically insignificant, we cannot make definitive claims. More data collection and research is needed to clarify these patterns and provide a more robust understanding.

Interestingly, after combining CRSV levels to create a nominal scale, the logistical regression statistical tests indicated a negative relationship between pre-conflict sexual violence and CRSV levels. While this relationship was also not statistically significant, it raises the question of whether there is a possibility that countries with lower levels of pre-conflict sexual violence might experience higher levels of CRSV. This could be due to the breakdown of social order and the lack of accountability, particularly from government forces, during conflict. However, as the null hypothesis could not be rejected, we cannot make these claims conclusively but encourage further research into this potential negative relationship.

Overall, the lack of statistically significant relationships prevented us from answering our research question about what explains the variability in CRSV across different conflicts in different countries. Nonetheless, this study provided valuable insights and highlighted opportunities for future research into this complex and sensitive topic. This also suggests that other factors—such as GDP of countries, the level of representation of women in different national armies, average income of women in different countries and the Gender Inequality Index, or other socio-economic variables—might better explain the variability in CRSV levels across countries.

Multiple limitations must be considered when examining this study and  interpreting the results of this study. Themajor limitation was lack of data, because of the the lack of data multiple countries were not included in teh final combined dataset as our CRSV was only from 1999-2011, hence countries that starte before or after that time frame had to be omitted.This led to our small the sample size of 130 countries may have been insufficient to detect more subtle relationships between pre-conflict sexual violence and CRSV. Additionally, the study was limited by the reliance on pre-conflict sexual violence data for only one year prior to the conflict, which may not have fully captured the broader context of sexual violence in these countries, as the continuum of violence might have stared even during the high tensions before conflict. Future studies could expand this timeframe to include data from multiple years leading up to the conflict.Another limitation was the absence of actual counts or percentages of preconflict sexual violence and only availability of the log transformation of the preconflict prevalence, which made interpretation more challenging, and might have skewed the detectness of linear relationship. Furthermore, the focus on intrastate CRSV perpetrated by government forces might havve excluded the possibility of considering CRSV committed by external actors, such as non-state armed groups or foreign intervention forces. This exclusion may have overlooked important dynamics affecting the overall levels of CRSV.

In conclusion, this paper explored the variability in conflict-related sexual violence (CRSV) across different countries, focusing on potential factors that contribute to its prevalence. The literature review emphasized the role of pre-conflict sexual violence as a possible predictor of CRSV, leading to the

hypothesis that countries with high levels of peacetime sexual violence would experience higher CRSV during conflict. However, both bivariate and multivariate analyses failed to reject the null hypothesis, challenging the original hypothesis. The small sample size of 130 countries may have contributed to high p-values, limiting the statistical significance of the results. Despite the lack of statistically significant relationships, this study offers valuable insights into the complexities of CRSV and underscores the need for more comprehensive research. The limitations of this study, particularly data constraints and the exclusion of CRSV perpetrated by non-state actors highlight the urgent need for more robust and diverse data collection in the field of women, peace, and security. Women continue to suffer from sexual violence during conflicts across the world, and understanding the dynamics of this violence is essential for prevention and intervention. Without a deeper, more nuanced understanding of CRSV, efforts to protect women and prevent sexual violence will remain insufficient.

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**VII. Appendix:**

**I. R studio code:**

```
#1 Filtering of Dataset of of CRSV committed by government from 1990 till 2011(Datasource Butler (US Dept States )) (CRSV Score)

SVGS <- read_dta("SVGSFdata_1999-2011.dta")

CRSV <- SVGS %>%

  dplyr::select(isoshnm, year, statedeptsvcode) %>%

  rename(Country = isoshnm, Year = year ) %>%

   group_by(Country) %>%  # Group the data by Country

   slice_max(order_by = statedeptsvcode, n = 1, with_ties = FALSE) %>% # Select the row with the highest CRSV value for each group, the first occurence in a country

   ungroup()

head(CRSV)



 #2 Filter for conflict years and first year of conflict years)

library(readxl)

#Filtering to only have the varaibles with active conflict, have Year conflict ID and Location

  filtered_data <- CRSV[SVAC$conflictyear == 1, c( "year", "conflictid","location"), drop = FALSE]

  filtered_data

Pre_Conflict_Year<- filtered_data %>%

  group_by(location) %>%

  summarize(first_year_of_conflict = min(year, na.rm = TRUE)) %>%

  mutate(Year0 = first_year_of_conflict - 1) %>%  #To identify the preconflict year

  mutate(location = toupper(location)) %>%

  rename( Country = location) %>%

   filter(!grepl(",", Country)) %>% #Filtering rows with two countries(So only intrastate countries)

  mutate(Country = case_when(
```

```
    Country == "BOSNIA-HERZEGOVINA" ~ "BOSNIA AND HERZEGOVINA",
    Country == "COTE D'IVOIRE" ~ "CÔTE D'IVOIRE",
     Country == "DEMOCRATIC REPUBLIC OF CONGO (ZAIRE)" ~ "DEMOCRATIC REPUBLIC OF
THE CONGO",
   Country == "IRAN" ~ "IRAN (ISLAMIC REPUBLIC OF)",
   Country == "LAOS" ~ "LAO PEOPLE'S DEMOCRATIC REPUBLIC",
   Country == "MACEDONIA" ~ "NORTH MACEDONIA",
    Country == "PAPUA NEW GUINEA" ~ "PAPUA",
   Country == "RUSSIA" ~ "RUSSIAN FEDERATION",
   Country == "YUGOSLAVIA" ~ "SERBIA",
   Country == "SYRIA" ~ "SYRIAN ARAB REPUBLIC",
   Country == "TANZANIA" ~ "UNITED REPUBLIC OF TANZANIA",
   Country == "VENEZUELA" ~ "VENEZUELA (BOLIVARIAN REPUBLIC OF)",
    TRUE ~ Country  # Keep all other entries unchanged
  ))


#3 Filter GBD dataset
SEV_SV <- read_csv("Edited_SEV_scalar_agestd_inj_homicide_sexual.csv")
head(SEV_SV)
Filtered_Sexual_Violence <- SEV_SV %>%
  dplyr::select(location_name, covariate_name_short,sex_label, year_id, val) %>%
  rename(Country = location_name, Year0 = year_id, sev_sv = val) %>%
  mutate(Country= toupper(Country)) %>%
  dplyr::select(-c(covariate_name_short))


#4 Combining both datasets:
```

```
Pre_Conflict_SV <- merge(Pre_Conflict_Year, Filtered_Sexual_Violence , by = c("Country", "Year0"), all.x = TRUE)

Final_Data <- merge(CRSV,Pre_Conflict_SV, by= c("Country"), all.x = TRUE) %>%

  na.omit()


#5 Descriptive_Statitics

dim(Final_Data)

nrow(Final_Data)#Numberofconflicts

names(Final_Data)#We'll change and distinguish Year.x and Year.y


#6 Data descriptives/univariate statistics

summary(Final_Data) #provides summary of ALL variables in data set: minimum, maximum, median, mean, NAs

summary(Final_Data$statedeptsvcode) #provides summary for variable presvote

summary(Final_Data$sev_sv)

Mode(Final_Data$statedeptsvcode)

Mode(Final_Data$sev_sv)


# 7 Barplot of distribution of Dependent Variable(CRSV) WHICH is an ordinal measure

Final_Data %>%

  ggplot(aes(x = statedeptsvcode)) +

  geom_bar()


#8:Dispersion:

range (Final_Data$sev_sv)

range (Final_Data$statedeptsvcode)
```

```
#standard deviation for a variable - suitable for interval and ratio level data
sd(Final_Data$sev_sv) #provides standard deviation for age variable

#9 Bivariate Anovatest
aov(statedeptsvcode ~ sev_sv, data = Final_Data)
results.aov <-aov(statedeptsvcode ~ sev_sv, data = Final_Data)
summary(results.aov) #shows F test results, p value.
aggregate(Final_Data$sev_sv,list(Final_Data$statedeptsvcode),FUN=mean)
etaSquared(results.aov)

# 10 ANOVA TEST for just female
Final_data1 <- Final_Data %>%
  filter(sex_label != "Male") #Filter out male
aov(statedeptsvcode ~ sev_sv, data = Final_data1)
results.aov <-aov(statedeptsvcode ~ sev_sv, data = Final_data1)
summary(results.aov) #shows F test results, p value.
aggregate(Final_data1$sev_sv,list(Final_data1$statedeptsvcode),FUN=mean)

# 11 Bivariate logistical regression:
Final_Data_nominal <- Final_Data %>%
  mutate(CRSV_nominal = case_when(
    statedeptsvcode == 0 ~ 0,    # Recode 0 as "no CRSV"
    statedeptsvcode >= 1 ~ 1        # Recode 1 and higher as "CRSV"
  ))
logtest <- glm( CRSV_nominal ~ sev_sv, family = "binomial", data= Final_Data_nominal)
```

summary(logtest)$coefficients

pR2(logtest)

#12 Multivariate Analysis: Logistical regression with two predictors.

logtest2 <- glm( CRSV_nominal ~ sev_sv + sex_label, family = "binomial", data= Final_Data_nominal)

summary(logtest2)$coefficients

pR2(logtest2)

## II. Tables created in Latex.

# Empirical Tables

Kampe Rushoka

December 2024

Table 1: Coding Rubric for Sexual Violence Committed by Security Forces according to Butler et.all

| Level | Description |
|---|---|
| 0 | No mention of 'rape,' 'sexual assault,' or 'sexual abuse' by security forces. |
| 1 | Isolated or alleged reports of 'rape,' 'sexual assault,' or 'sexual abuse' by security forces. |
| 2 | 'Reports' or indications that SV occurred 'sometimes,' 'continued to be a problem,' or 'remained an issue.' |
| 3 | 'Numerous reports' of SV described as 'routine,' 'common,' 'widespread,' or 'systematic,' including its use as a tool of torture to extract information, intimidate, or punish targeted groups (e.g., detainees, opposition members, or ethnic minorities). |
| 4 | SV was used as 'a tool of war' or 'a systematic weapon of war.' |

| Variable | Estimate | Std. Error | z value | Pr(¿—z—) |
|---|---|---|---|---|
| (Intercept) | 2.6289 | 0.5086 | 5.1689 | $2.36 \times 10^{-7}$ |
| sev_sv | -0.6657 | 1.1187 | -0.5950 | 0.5518 |

Table 2: Regression Results 1: Estimates, Standard Errors, z Values, and p-Values

| Variable | Eta Squared | Partial Eta Squared |
|---|---|---|
| sev_sv | 0.007094883 | 0.007094883 |

Table 3: Eta Squared Results

| Variable | Estimate | Std. Error | z value | p-values |
|---|---|---|---|---|
| (Intercept) | 3.0624 | 0.9640 | 3.1768 | 0.0015 |
| sev_sv | -1.2497 | 1.5353 | -0.8140 | 0.4156 |
| sex_labelMale | -0.4618 | 0.8548 | -0.5403 | 0.5890 |

Table 4: Regression Results 2: Estimates, Standard Errors, z Values, and p-Values



## III. Latex Code for tables:

```
\documentclass{article}

\usepackage{graphicx} % Required for inserting images

\title{Empirical Tables}

\author{Kampe Rushoka}

\date{December 2024}

\begin{document}

\maketitle

\begin{table}[h!]
```

```
\centering

\caption{Coding Rubric for Sexual Violence Committed by Security Forces according to Butler et.all}

\begin{tabular}{|c|p{12cm}|}

\hline

\textbf{Level} & \textbf{Description} \\ \hline

0 & No mention of ‘rape,’ ‘sexual assault,’ or ‘sexual abuse’ by security forces. \\ \hline

1 & Isolated or alleged reports of ‘rape,’ ‘sexual assault,’ or ‘sexual abuse’ by security forces. \\ \hline

2 & ‘Reports’ or indications that SV occurred ‘sometimes,’ ‘continued to be a problem,’ or ‘remained an issue.’ \\ \hline

3 & ‘Numerous reports’ of SV described as ‘routine,’ ‘common,’ ‘widespread,’ or ‘systematic,’ including its use as a tool of torture to extract information, intimidate, or punish targeted groups (e.g., detainees, opposition members, or ethnic minorities). \\ \hline

4 & SV was used as ‘a tool of war’ or ‘a systematic weapon of war.’ \\ \hline

\end{tabular}

\label{tab:sv_rubric}

\end{table}



\begin{table}[h!]

\centering

\begin{tabular}{lcccc}

\hline

\textbf{Variable} & \textbf{Estimate} & \textbf{Std. Error} & \textbf{z value} & \textbf{Pr(>|z|)} \\

\hline

(Intercept)       & 2.6289 & 0.5086 & 5.1689  & $2.36 \times 10^{-7}$ \\

sev\_sv           & -0.6657 & 1.1187 & -0.5950 & 0.5518 \\

\hline

\end{tabular}

\caption{Regression Results 1: Estimates, Standard Errors, z Values, and p-Values}

\label{tab:regression_results1}

\end{table}
```

```
\begin{table}[ht]
\centering
\begin{tabular}{lcc}
\hline
\textbf{Variable} & \textbf{Eta Squared} & \textbf{Partial Eta Squared} \\
\hline
sev\_sv & 0.007094883 & 0.007094883 \\
\hline
\end{tabular}
\caption{Eta Squared Results}
\label{tab:eta_squared}
\end{table}

\begin{table}[h!]
\centering
\begin{tabular}{lcccc}
\hline
\textbf{Variable}     & \textbf{Estimate} & \textbf{Std. Error} & \textbf{z value} & \textbf{p-values} \\
\hline
(Intercept)          & 3.0624  & 0.9640  & 3.1768  & 0.0015 \\
sev\_sv              & -1.2497 & 1.5353  & -0.8140 & 0.4156 \\
sex\_labelMale       & -0.4618 & 0.8548  & -0.5403 & 0.5890 \\
\hline
\end{tabular}
\caption{Regression Results 2: Estimates, Standard Errors, z Values, and p-Values}
\label{tab:regression_results2}
\end{table}
```

```
\end{document}
```